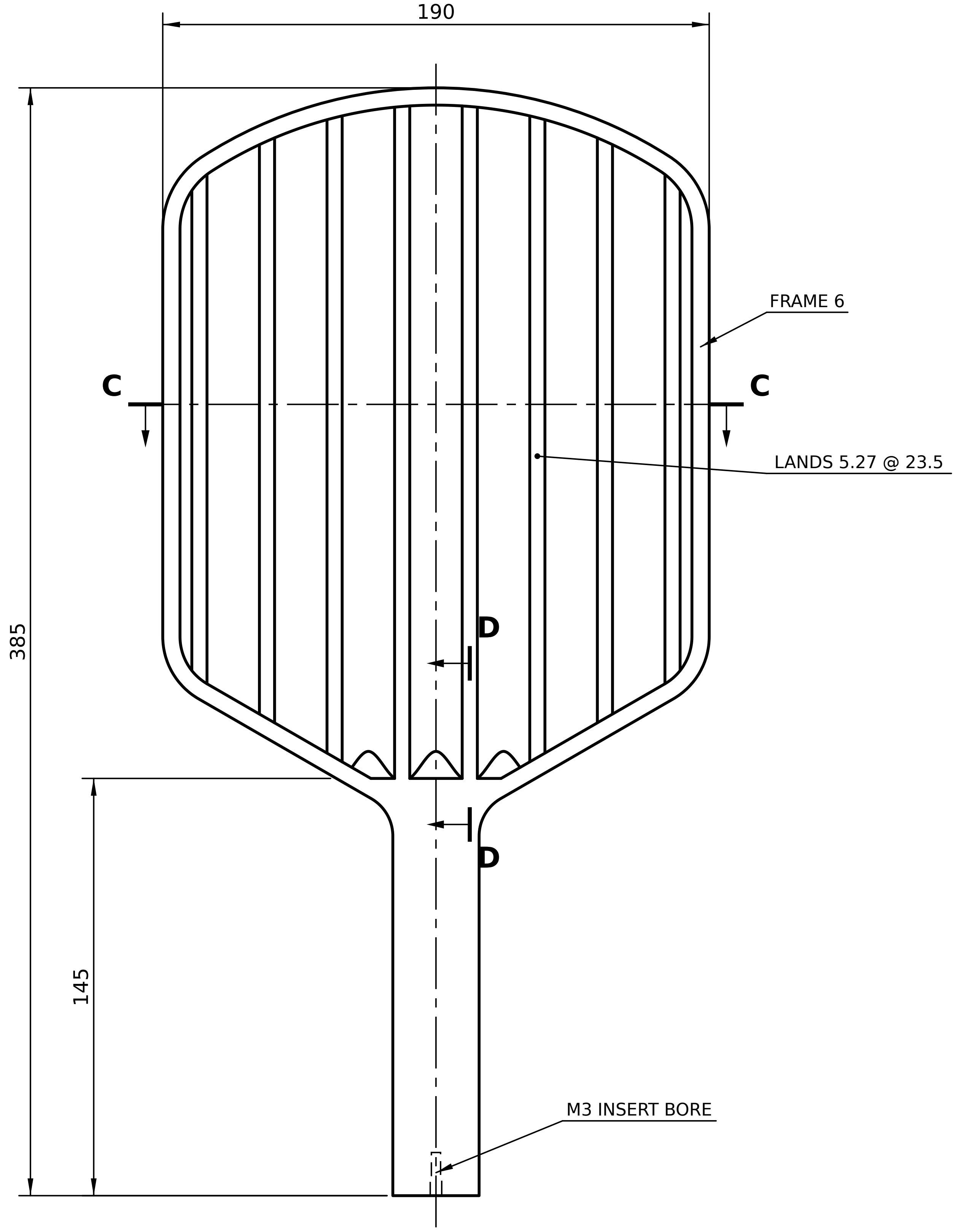
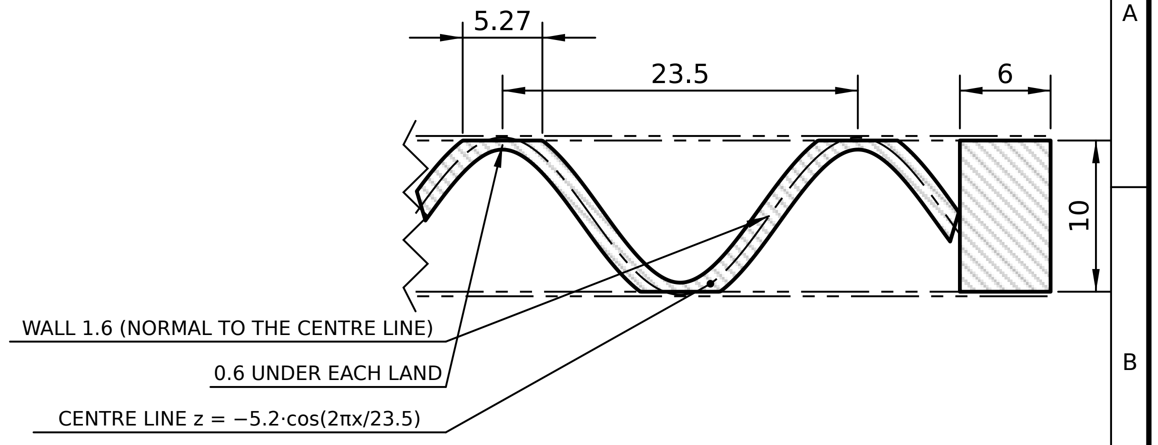


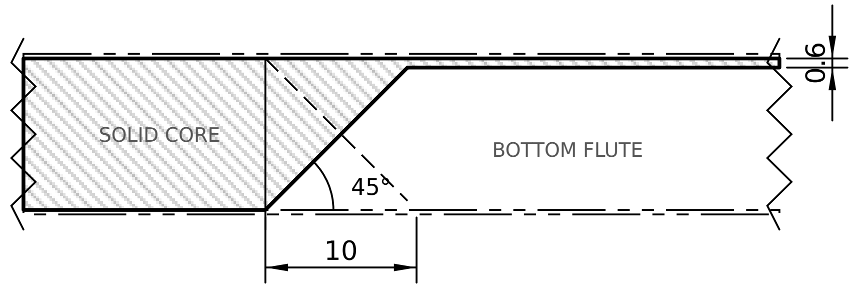
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH


5.27
23.5
6
10
WALL 1.6 (NORMAL TO THE CENTRE LINE)
0.6 UNDER EACH LAND
CENTRE LINE z = −5.2·cos(2πx/23.5)

SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON A LAND (x = 11.75), RAMP FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: CARBON FIBRE / EPOXY, MOULDED IN ONE PIECE IN THE SHEET-4 MOULD (CHOPPED FIBRE 10–25 LONG, ≈ 40 % BY VOLUME, ρ ≈ 1.5 g/cm³). THE PRINTED STL (NOTE 8) IS A TRIAL PIECE / PRINTED ALTERNATIVE.
2. THICKNESS 10, SET BY THE MOULD: THE LID FLANGE SEATS ON THE TRAY RIM WITH THE PLUG FACE 10 ABOVE THE POCKET FLOOR (SHEET 4). BOTH FACES ARE BOND SURFACES.
3. WAVE RIB λ23.5 (8 WAVES ACROSS, CRESTS SYMMETRIC ABOUT THE CENTRE LINE), WALL 1.6, CENTRE-LINE AMPLITUDE 5.2; CRESTS CUT FLAT = LANDS 5.27 ON BOTH FACES.
4. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE ENDS IN A 45° RAMP, 10 LONG, LONGEST AT ITS OPEN FACE (D–D), SO THE MOULD TEETH PULL STRAIGHT OUT.
5. AFTER MOULDING DRILL Ø4 × 7 + Ø3.2 × 15 IN THE BUTT FACE AND BOND AN M3 BRASS INSERT (EPOXY) FOR THE GRIP SCREW – SEE PB-GR-002.
6. BONDING THE SKINS: ABRADE (P120) AND DEGREASE BOTH FACES, EPOXY, PRESS FLAT BETWEEN PLATENS OR IN A VACUUM BAG.
7. PRINT FILE output/rib_support/hybrid_rib_support.stl (TRIAL PIECE: CHECKS THE MOULD CLOSES): FLAT, BED ≥ 391 × 196.
8. MASS 2[illegible]9 g. y FROM THE GRIP BUTT FACE (y = 0); THE RIB SUPPORT STARTS AT y = 5.

RIB SUPPORT – HYBRID

RIB SUPPORT (CORE) · MOULDED CARBON, ONE PIECE · 385 × 190 × 10 mm

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-11-09		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	F	3 / 4	A3	1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
◎▷	CARBON / EPOXY (NOTE 1)	ISSUED FOR REVIEW